JASON ZHI LIANG, Ph.D.

Principal Research Scientist | Evolutionary AI | Self-Improving Agentic Systems

San Francisco, CA

Contact: jasonzliang@utexas.edu | (510) 378-0709

Links: Homepage · LinkedIn · Google Scholar · GitHub

Research Summary

Principal Research Scientist at Cognizant AI Lab, leading work on **automated AI research**, **open-ended learning**, and **self-improving agentic systems**, built on a decade of large-scale neuroevolution and evolutionary optimization for deep learning. My focus: automating the AI-research loop, from designing architectures and training recipes to knowledge synthesis, with minimal human intervention, guided by two convictions: a system's founding values compound over its lifetime, and open-endedness and creativity must continue at inference time, not just during training.

My work spans three layers of the AI stack: **CoDeepNEAT** (evolutionary neural architecture search; multitask extension, GECCO 2018), **EPBT** (loss-function and hyperparameter evolution, GECCO 2021), and **Caesar** (Deep Research agent, presented at ICML 2026 Human-AI Co-Creativity Workshop), with the same principle recurring at each level: evolve and select architectures and training recipes below, then critique and refine an agent's own reasoning above. Named inventor on 5 issued U.S. patents in evolutionary AI and neural architecture search.

Technical Expertise

- **Evolutionary optimization & architecture search:** neuroevolution, neural architecture search, population-based training, multi-task learning, evolutionary loss/hyperparameter optimization.
- **Agentic & self-improving systems:** deep-research agents, multi-agent orchestration, inference-time exploration, values-guided self-improvement, LLM-as-judge evaluation.
- **Distributed systems for search:** asynchronous evaluation and scheduling, cluster-scale parallel search.
- **Languages & frameworks:** Python, TensorFlow and Keras, LLM APIs (Gemini, Claude, GPT), agentic frameworks including Claude Code.

Professional Experience

Cognizant AI Lab | Principal Research Scientist

San Francisco, CA | Dec 2018 – Present | Lead Researcher driving the roadmap for the lab's automated AI research effort.

1. Autonomous Self-Improvement Systems

- **Values-Based Self-Improvement Framework:** Autonomous self-improvement system built on Claude Code, organized around studying three principles: (1) the founding values and biases seeded into the agent; (2) invariants against reward hacking that the agent cannot overwrite; (3) how much of the agent's own instructions and values it may rewrite. Early-stage research into how starting conditions and self-modification interact and compound to shape what an agentic system builds. On the variable radius circle packing task the agent wrote a unique solver that matched the best-known value at N=26 for under $20 of compute to first tie, with no web assistance. At N=27 the same solver beat the existing record, reaching 2.685978684198. The agent later discovered another solver that took 21 more. Both are listed on Packomania as of 11 August 2026.

2. Caesar: Creative Reasoning & Knowledge Synthesis

- **Automated Research via Inference-Time Self-Refinement:** Designed and implemented "Caesar," an autonomous Deep Research agent. Scaled inference-time compute using an adversarial refinement loop that actively critiques internal drafts, generates orthogonal queries to target weaknesses, and consolidates findings via a generative merge. **Outperforms the strongest of three frontier deep-research agents by 13–23% on a blinded, LLM-judged creative-synthesis benchmark (Cliff's $\delta \geq 0.76$).**
- **Graph-Based Associative Reasoning:** Replaced static RAG retrieval with a "Perceive-Think-Act" loop over a dynamic knowledge graph, enabling discovery of non-obvious cross-disciplinary connections that linear retrieval tends to miss.
- **Open-Ended Exploration Policy:** Designed an active information-foraging policy that detects stagnation in memory and graph structures, autonomously executing strategic backtracking to maximize information gain over long horizons. **Ablation: reducing exploration budget from 1000 to 250 iterations measurably degrades long-form answer quality, validating that deep topological traversal is essential for creative synthesis.**

3. Multi-Agent Systems & Large-Scale Evolutionary Optimization

- **Multi-Agent Code Generation:** Developed a **Finite State Machine (FSM)** framework to coordinate teams of specialized coding agents. Applied evolutionary algorithms to optimize the *composition* of these teams, evaluated on SciCode and HumanEval+ code-generation benchmarks.
- **Evolutionary Population-Based Training (EPBT):** Pioneered a population-based approach that evolves loss functions and hyperparameters jointly with weights via population-based weight sharing. **Outperforms the PBT baseline by 1.3pp on CIFAR-10 ResNet-32 (92.79% vs 91.53%) while exploring 520 candidate loss functions at the cost of 40 trainings, a 13× reduction** (GECCO 2021; U.S. patent pending).
- **Code Repository Analysis:** Created "Code Archaeologist," an agentic system capable of reasoning over repository histories (via Git/LFS) to detect architectural patterns and technical debt.

Sentient Technologies | Research Scientist (prev. Research Intern)

San Francisco, CA | Dec 2015 – Nov 2018

- **Distributed Evolutionary Architecture Search:** Scaled **CoDeepNEAT** (evolutionary neural architecture search, developed during my UT Austin PhD) to GPU clusters. Validated that population-based search scales with parallel compute; established asynchronous evolutionary optimization at scale. Productionized for image captioning with Sentient's product team.

- **Multi-Task Learning:** Achieved then state-of-the-art results on the **20-task Omniglot multi-task learning benchmark (88% accuracy, +21pp over Soft Ordering at 67%)** by evolving modular topologies with cross-task weight sharing.

The University of Texas at Austin | Research Assistant / Ph.D. Candidate

Austin, TX | Sep 2013 – Dec 2018

- **Bilevel Meta-Evolution for Control:** Developed **MEA (Meta-Evolutionary Algorithm)**, a bilevel framework that evolves the hyperparameters of neuroevolution; the meta-level objective is non-convex and non-differentiable, so gradient methods do not apply there. Applied to continuous-control tasks including helicopter hovering and double pole balancing.

Selected Publications & Patents

Full publication list available on Google Scholar

Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis (2026) *Jason Liang, Elliot Meyerson, Risto Miikkulainen*. arXiv:2604.20855. Presented at the **Human-AI Co-Creativity Workshop at ICML 2026**.

- Autonomous research agent that scales inference-time compute through an adversarial refinement loop over a dynamic knowledge graph, outperforming the strongest of three frontier deep-research agents on a blinded, LLM-judged creative-synthesis benchmark.

Asynchronous Evolution of Deep Neural Network Architectures (2024) *Jason Liang, Hormoz Shahrzad, Risto Miikkulainen*, Applied Soft Computing, Vol. 152.

- Techniques for asynchronous population-based optimization at scale, essential for efficient training on massive distributed clusters.

Regularized Evolutionary Population-Based Training (2021) *Jason Liang, Santiago Gonzalez, Hormoz Shahrzad, and Risto Miikkulainen*, GECCO 2021.

- Method to dynamically optimize loss functions during training, improving convergence speed and generalization.

Evolutionary Neural AutoML for Deep Learning (2019) *Jason Liang, Elliot Meyerson, Babak Hodjat, et al.*, GECCO 2019.

- Evolutionary AutoML that jointly optimizes architecture, hyperparameters, and network size, not hyperparameters alone.

Evolutionary Architecture Search for Deep Multitask Networks (2018) *Jason Liang, Elliot Meyerson, and Risto Miikkulainen*, GECCO 2018.

- Extends CoDeepNEAT to multitask architecture search, automatically discovering deep multitask networks that outperform human-designed baselines.

Evolving Deep Neural Networks (2017/2019) *Risto Miikkulainen, Jason Liang, Elliot Meyerson, et al.*, Artificial Intelligence in the Age of Neural Networks and Brain Computing.

- Book chapter surveying neuroevolutionary methods for deep learning; introduces CoDeepNEAT, the coevolutionary neural architecture search algorithm I invented (first named inventor, U.S. 11,003,994 and 11,250,328), with applications to language modeling, image captioning, and multi-task learning.

Evolutionary Bilevel Optimization for Complex Control Tasks (2015) *Jason Zhi Liang, Risto Miikkulainen*, GECCO 2015.

- Bilevel evolutionary optimization of neuroevolution hyperparameters, applied to continuous-control tasks (helicopter hovering, double pole balancing).

Patents (Issued, U.S.):

- **US Patent 12,282,845:** Multi-objective Coevolution of Deep Neural Network Architectures (2025).
- **US Patent 11,507,844:** Asynchronous Evaluation Strategy for Evolution of Deep Neural Networks (2022).
- **US Patent 11,250,328:** Cooperative Evolution of Deep Neural Network Structures (2022).
- **US Patent 11,030,529:** Evolution of Architectures for Multitask Neural Networks (2021).
- **US Patent 11,003,994:** Evolutionary Architectures for Evolution of Deep Neural Networks (2021).

Patents (Pending, U.S.):

- **App. 17/389,961:** Regularized Evolutionary Population-Based Training (2023).
- **App. 17/064,706:** Sharing Meta-Learning Methods Among Multiple Private Data Sets (2022).

Awards & Honors

- **BEACON Research Grant** (2015), BEACON Center for the Study of Evolution in Action (an NSF Science & Technology Center).

Professional Activities

- **ICML Silver Reviewer**, International Conference on Machine Learning (ICML)
- **Reviewer**, Genetic and Evolutionary Computation Conference (GECCO)

Education

Ph.D. in Computer Science · *Neuroevolution & Deep Learning*
The University of Texas at Austin · Sep 2013 – Dec 2018 · Advisor: Risto Miikkulainen
Dissertation: *Evolutionary Neural Architecture Search for Deep Learning*

B.S. in Electrical Engineering and Computer Science
University of California, Berkeley · Sep 2009 – May 2013